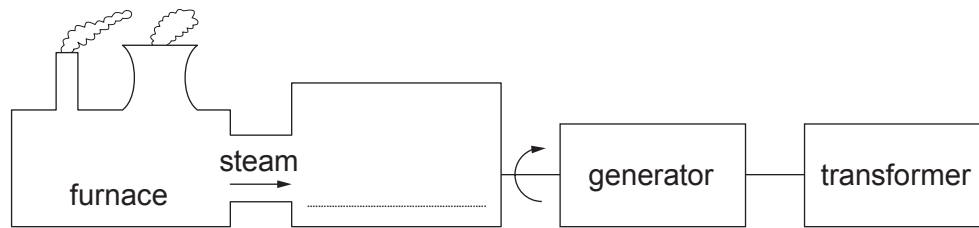


2. The diagram below shows the main parts of a thermal power station that uses coal to produce electricity.



- (a) (i) **Complete the labelling on the diagram.** [1]
 (ii) Draw **one** line from each part of the power station to the energy change it produces. [2]

furnace	kinetic energy to electrical energy
generator	chemical energy to electrical energy
	chemical energy to thermal (heat) energy

- (b) Some energy values for this power station are listed below.

Input energy = 15 000 kJ
 Heat energy = 10 500 kJ
 Electrical energy = 4 500 kJ

Use this information and the equation below to calculate the % efficiency of the power station. [2]

$$\% \text{ efficiency} = \frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$$

% efficiency =

